最长三段连续子序列和（不要求上升）-板子

徐版：

```C++
#include<bits/stdc++.h>
using namespace std;

void run(){
    int n;cin>>n;
    vector<int> a(n+1);
    vector<int> ans1(n+1,-1e9);
    vector<int> ans2(n+1,-1e9);
    for(int i=1;i<=n;i++) cin>>a[i];
    ans1[1] = a[1];
    ans2[n] = a[n];
    int maxn = a[1];
    int tmp = a[1];
    for(int i=2;i<=n;i++){
        int t = max(a[i]+tmp,a[i]);
        maxn = max({tmp,maxn,t});
        ans1[i] = maxn;
    }
    tmp = a[n];
    maxn = a[n];
    for(int i=n-1;i>=1;i--){
        int t = max(a[i]+tmp,a[i]);
        maxn = max({tmp,maxn,t});
        ans2[i] = maxn;
   }
    maxn = -1e9;
    for(int i=3;i<=n-2;i++){
        maxn = max((ans1[i-2]+ans2[i+2]+a[i]),maxn);
    }
    cout<<maxn<<"\n";
    return;
}

signed main(){
    ios::sync_with_stdio(0);
    cin.tie(0);
    cout.tie(0);
    int t;cin>>t;
    while(t--) run();
}
```

wyr版：

C++

```cpp
#include<bits/stdc++.h>
using namespace std;
using ll = long long; const ll MIN = -1e16;
//vector<int> ans(M, 0);

void work(){
    ll n;
    cin >> n;
    vector<int> a(n);
    for(int i=0; i<n; ++i){
        cin >> a[i];
    }
    ll mx1 = MIN, ans = MIN, tmp1 = 0, tmp2 = 0;
    vector<ll> mx2(n+1, MIN);
    for(int j=n-1; j>=4; --j){
        ll cur2 = a[j];
        tmp2 = max(tmp2 + cur2, cur2);
        mx2[j] = max(mx2[j+1], tmp2); // j~n-1 的最大子段和
    }
        for(int i=2; i<=n-3; ++i){
//        mx2 = MIN;
        ll cur1 = a[i-2];
        tmp1 = max(tmp1+cur1, cur1);
        mx1 = max(mx1, tmp1);
        //        mx2 = max(mx2, tmp2[i+2]);
                ll cur = mx1 + a[i] + mx2[i+2];
        ans = max(cur, ans);
//        cout << "mx1:" << mx1 << " a[i]:" << a[i] << " mx2:" << mx2 << " ans:"  << ans << '\n';     }
    cout << ans << '\n';
}

signed main(){
    ios::sync_with_stdio(0);
    cin.tie(0);
        int T=1;
    cin >> T;
    while(T--) work();
}

/*
3
5
-1 -2 5 6 1
6
5 4 1 3 6 4
7
1 -2 3 4 2 -1 -7

5
16
```

```
6
*/
```